

Introduction to data visualisation

Data Visualisation

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Week 1

Course structure

- 10 weeks
- all lectures and labs on campus

Lecture schedule:

•Venue: RHB 309, Richard Hoggart Building

•Lab schedule:

Venue: WB 219, Whitehead Building [PC Lab]

Time: Thursday 18-19:30PM

Kindly check your time daily for updates

Assessment

100% coursework

Coursework

Group survey project [50%]

- Survey design (group)
Formative
- Survey report (group) [50%]

Individual project [50%]

Timeline

- see VLE...

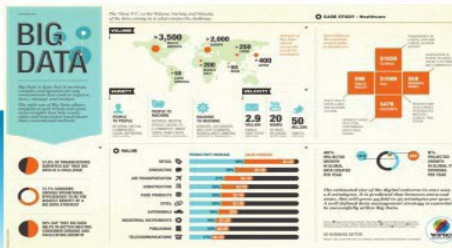
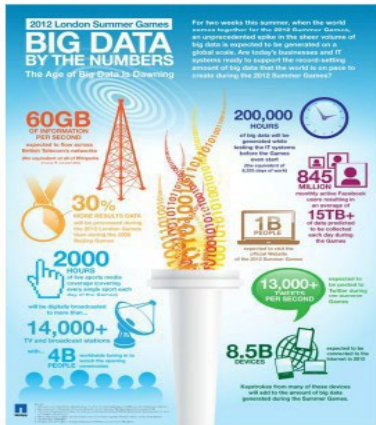
Questions ?

Data is generated
every day and everywhere



Data in ever increasing sizes ⇒ need an effective way to understand them

Age of Big Data



Companies in all sectors have at least 100 terabytes of stored data in the United States; many have more than 1 petabyte



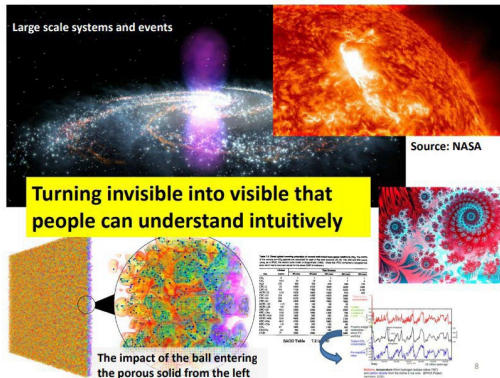
- 1 Storage data by sector derived from IDC.
- 2 Firm data split into sectors, when needed, using employment
- 3 The particularly large number of firms in manufacturing and health care provider sectors make the available storage per company much smaller.

SOURCE: IDC; US Bureau of Labor Statistics; McKinsey Global Institute analysis

What is Visualisation?

- Data visualisation leverages various visual tools such as charts, graphs, maps, and infographics in transforming complex data into intuitive and actionable form.
- Data visualisation helps bridge the gap between data analysis and storytelling, making it easier understand and interpret information more effectively.
- Visualisation transforms data into images that effectively and accurately represent information about the data.

Tool to enable a User insight into Data



Let's collect some data!

The data visualisation process

1. Problem statement (define concepts and questions)
2. Data collection
3. Data cleaning and pre-processing
4. Exploratory DV (looking for patterns and trends)
5. Explanatory DV (communicating findings)

Problem statement

- Who is taking the DV module this year?
- What programmes are DV students following?
- What is the gender diversity in DV?
- What is the gender diversity in DV by programme?

*could be many other factors to consider,
but always have to be selective*

Variables

Variable 1 preferred pronoun

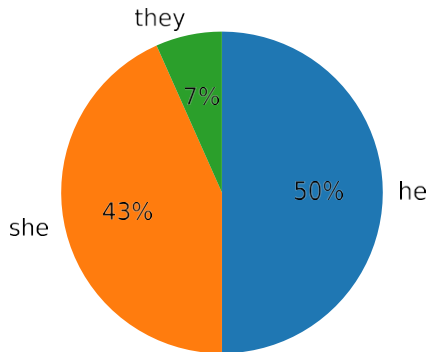
- she/her, he/him, they/their, name only...
- less intrusive than asking about gender directly

Variable 2 programme of study

- creative computing, business computing, computer science, data science...

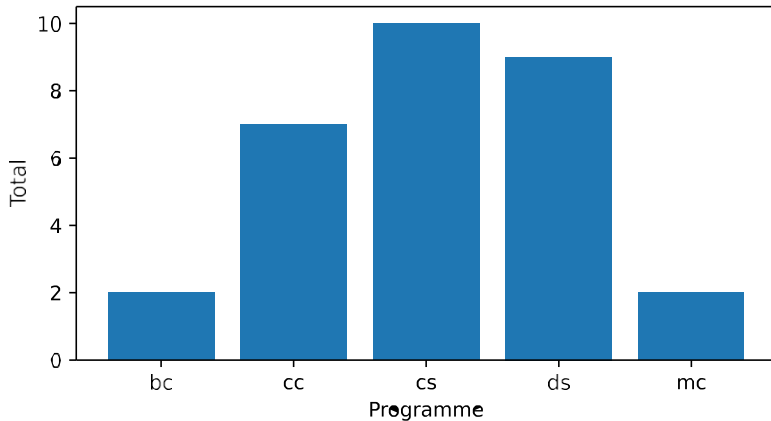
Visualisation: Pronouns pie chart

Pronouns used by Data Visualisaion students
(example data)



Visualisation: Programmes bar chart

Total number of Data Visualisation students
on each degree programme (example data)



Data

- Data are raw facts, figures, observations, or information collected or stored in various forms.
- It can be in the form of numbers, text, images, audio, video, or any other representation of facts or events.



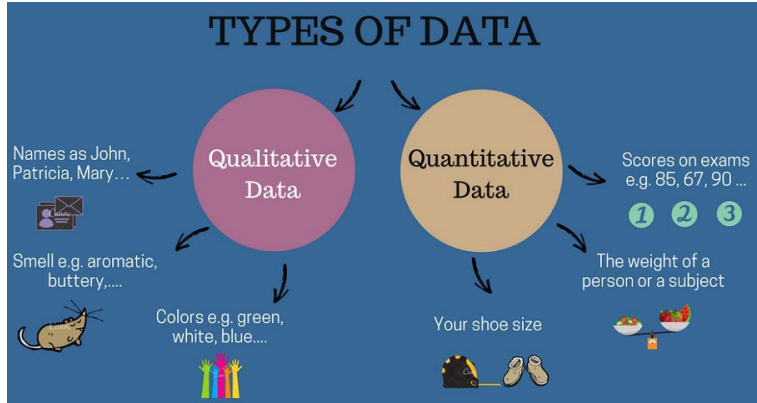
Data-“facts”

Quantitative

measur ement s of *quantity*

Qualitative

measur ement s of *quality*



is data enough?

How do we make data useful?

Data → Information

How do we make data useful?

Data → Information

Visualisation

- using graphics to explore and understand data (use human visual capacities)

Data mining

- searching for patterns and trends

Machine learning

- build models, test theories, predict the future

The information problem

- We are **data rich**
- But **information poor**

DRIP syndrome

Data & information

data $\neq$ information

Data & information

data $\neq$ information

Data Vis: data $\rightarrow$ information

(using power of human visual processing)

Data visualisation

*making **visible** the patterns and structures in data
using graphs, tables, diagrams, maps etc.*

Amplify cognition

The purpose of data visualisation

To reveal things that are

- important
- meaningful
- useful

Understanding

This is a well rich and inter-disciplinary area
that combines
knowledge from various disciplines

Evolution of Visualisation Research

- From direct visualisation to derived information visualization.
- From simple data to more complex ones.
- From represent the data with fidelity to reveal new findings.
- From scientific visualisation to information visualization, bio-visualisation, geographical data visualization, and beyond.

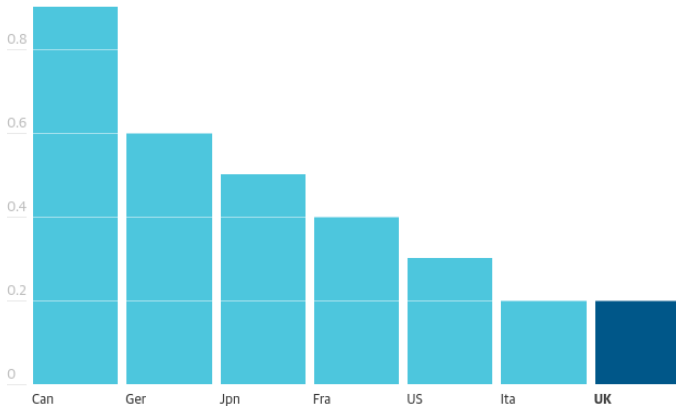
Information vs scientific visualisation

Information visualisation

- quantitative data representing abstract concepts, processes and behaviours
 - e.g. share prices, child literacy, sales, happiness...
- focus on understanding social, psychological, economic phenomena
- high-dimensional ($>>4$),
- Data is discrete in the nature – Examples include financial, marketing, HR, statistical, social media, political,
- Feature are not well-defined, the typical analysis tasks including finding
- patterns, clusters, voids, outliers

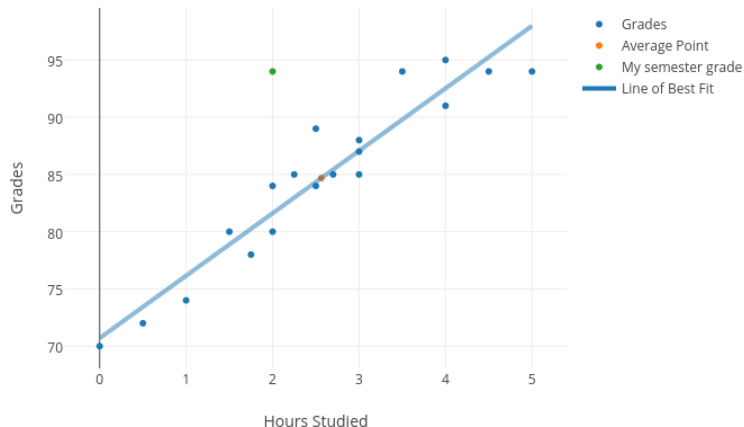
Information: Econ omy

Quarter-on-quarter GDP growth rates for G7 countries in the first three months of 2017, percentage



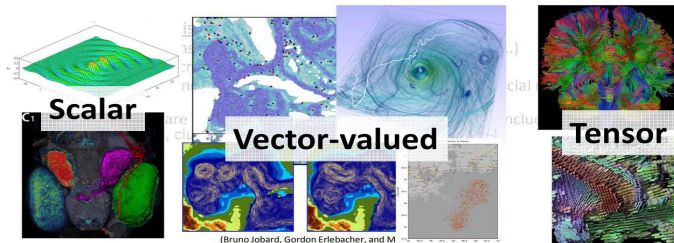
Information: Academic achievement

Grades in Comparison to Study Time

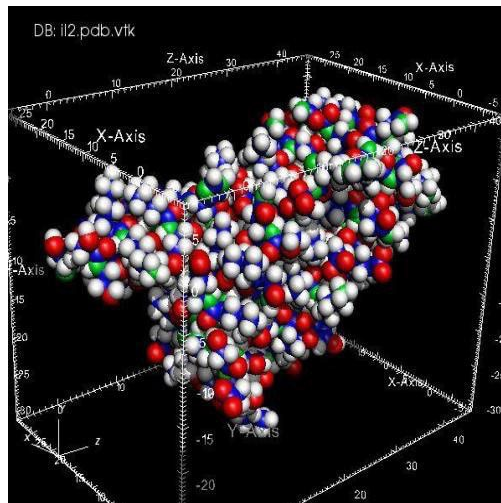


Scientific: is mostly concerned with

- Data defined in physical space, i.e. spatio-temporal data (2~4 dimensions)
 - Data describes continuous events in continuous space, however, the representation is discrete (i.e. sampled data)
 - Examples include simulation and measurement data from physics, chemistry, geo-science, medical-biological, climate, oceanography, energy
- (quantitative data representing physical objects)
Features are well-defined



Scientific: Molecular structure



Information vs scientific visualisation

Similarities

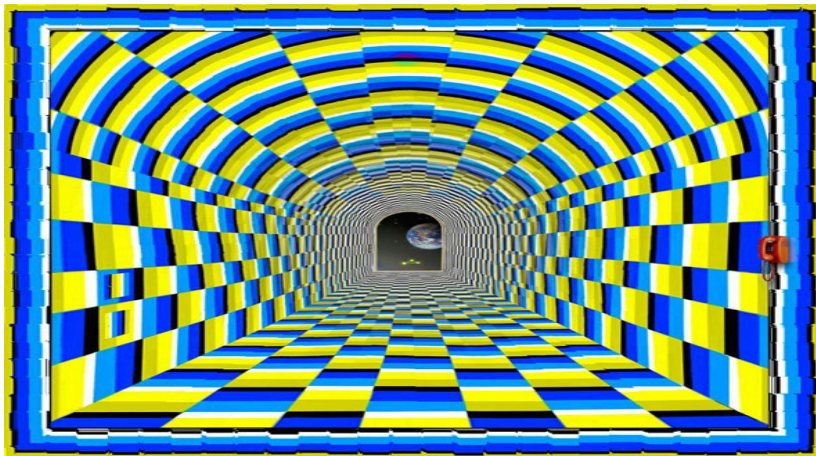
- use visual modality
- represent data in ways our brain can comprehend

Shared objectives

- amplify cognition
- create and communicate knowledge

Visual perception and design

*Visual displays provide the highest bandwidth channel from the computer to the human. Indeed, we acquire more information through vision than through all of the other senses combined.
(Ware 2012, p. 2)*



Semiotics of graphics

study of visual **symbols** and how they convey meaning

- *classical view* (Saussure)
- all symbols are arbitrary
- must be learnt
- no system of representation can be better than any other
- scientific view*
- the visual system has particular properties (product of evolution)
- different forms of representation are more closely aligned with our perception mechanisms
- there is some grounds for determining better/worse forms of visual communication

Sensory vs arbitrary symbols

sensory “aspects of visualisations that derive their expressive power from their ability to use the perceptual processing power of the brain without learning”

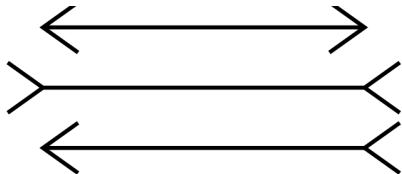
arbitrary “aspects of representation that must be learned, because the representations have no perceptual basis”

Ware (2012, p. 9)

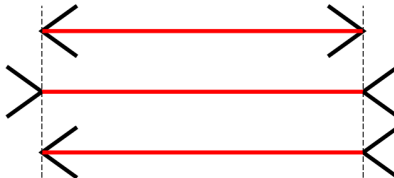
Sensory symbols

- well matched to the early stages of neural processing
- tend to be stable across individuals, cultures, and time
- representations can be understood without training

Müller-Lyer illusion



Source: Fibonacci / CC BY-SA



Source: Fibonacci / CC BY-SA

Gesturing reduces effect of a classic optical illusion

Arbitrary symbols

- derive power from culture
- dependent on individual cultural knowledge



Significance of symbolic, iconic and indexical relations in British road signs

Data visualisation

Exploratory

- helps discover trends and patterns
- interactive
- quick and dirty
- early stages of research

Explanatory

- conveys a clear message
- makes a point or answers a question
- polished, publication/presentation ready
- final stages of research

Uses for data visualisation

Think of DV examples in each context

1. analysis
2. communication
3. monitoring
4. planning

Value of data visualisation

- industry demand for data science skills
- useful in many areas of work and study
- practical data manipulation skills
- experience of research
- useful for third-year projects
- visualisation is a powerful means of communication, and making a difference

Research activities

Search online for examples of data visualisation.

In the forum, post the link to a good example you have found. Introduce the example and summarise the key message being communicated. Explain why you thought it an interesting and successful example of visualising data.

Recommended reading

General data visualisation

- A. Kirk (2016). *Data visualisation: a handbook for data driven design*. London: Sage Publications
- S. Few (2009). *Now You See It: Simple Visualization Techniques for Quantitative Analysis*. 1st ed. Oakland, CA, USA: Analytics Press

More theoretical

- G. J. Myatt and W. P. Johnson (2014). *Making Sense of Data I: A Practical Guide to Exploratory Data Analysis and Data Mining*. 2nd ed. Hoboken, NJ, US: John Wiley & Sons, Incorporated
- G. J. Myatt and W. P. Johnson (2009). *Making Sense of Data II: A Practical Guide to Data Visualization, Advanced Data Mining Methods, and Applications*. Hoboken, NJ, US: John Wiley & Sons, Incorporated

Technical and coding

- W. Mckinney (2017). *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*. Sebastopol, US: O'Reilly Media, Incorporated